Battery-Aware Multi-Sensor Exploration in Unseen Waters

IMSEL Research Group

CSCE 775: Deep Reinforcement Learning

***Abstract*—Autonomous Surface Vehicles (ASVs) deployed for long-duration monitoring face a critical trade-off between maximizing information gain and conserving limited battery life. High-fidelity sensors like Radar and Lidar provide rich data but consume significant power, while environmental factors like currents and waves impose varying energy costs on propulsion. This proposal studies an environment-aware reinforcement learning policy that jointly optimizes motion control and sensor duty-cycling. By conditioning the policy on inferred water-body types—such as lake, river, or coast—the agent learns distinct strategies to maximize coverage efficiency. We compare the proposed method against standard frontier exploration, rule-based coverage, and energy-unaware RL baselines, evaluating performance in terms of area mapped per Joule and mission success rate under diverse hydrodynamic conditions.**

I. Introduction

Autonomous Surface Vehicles (ASVs) are essential for scalable marine monitoring tasks such as bathymetry, pollution tracking, and border patrol. However, the operational range of these vehicles is strictly limited by battery capacity. High-fidelity perception using sensors like Radar and Lidar is crucial for safe navigation and detailed mapping, but these active sensors deplete energy rapidly. Furthermore, the energy cost of propulsion varies drastically across different water environments; moving against a strong river current consumes far more energy than navigating a calm lake.

This proposal asks a simple question: can an ASV learn to balance its sensor usage and motion strategies to maximize exploration coverage without running out of power? We propose a reinforcement learning (RL) policy that jointly controls vehicle velocity and sensor duty-cycling, explicitly conditioned on the water-body type ('lake', 'river', 'coast'). The core hypothesis is that an agent aware of both its internal energy state and external environmental dynamics will achieve higher information gain per unit of energy than standard energy-blind exploration methods or fixed-schedule sensor baselines.

The method builds on literature in active perception and energy-aware robotics [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [?], [1]. Prior work typically separates path planning from power management or assumes a static environment. Our contribution is a unified policy that treats sensor activation as a learned action and adapts its behavior to specific hydrodynamic contexts (Fig. effig:method-overview), with explicit comparison against related methods (Fig. effig:rw-matrix).

II. Approach

We formulate the autonomous exploration problem as a Partially Observable Markov Decision Process (POMDP), solved via Deep Reinforcement Learning (DRL). The core mechanism is the closed-loop interaction shown in Figure 1: the agent perceives the environment, acts, and learns from the consequences.

At each time step t, the environment presents a state s_t, consisting of fused sensor observations (IR/RGB/Radar), a local occupancy grid (belief map), the remaining battery state of charge E_t, and the water-body classification w_{type} (e.g., river current vs. static lake). The policy network $\pi_\theta(a_t|s_t)$ processes this state to output an action a_t. Crucially, this action space is hybrid, controlling both the USV's motion (velocity v, turn rate ω) and its perception hardware (duty-cycling sensors on/off to save power).

Upon execution, the physical environment processes the action—applying hydrodynamic drag, currents, and wind—to produce a new state s_{t+1} (new position, updated battery level). Simultaneously, a reward signal r_t is computed to quantify performance:

$$r_t = \lambda_1 \cdot \Delta\text{InfoGain} - \lambda_2 \cdot \text{EnergyCost}(a_t, w_{type}) - \lambda_3 \cdot \text{CollisionPenalty} \quad (1)$$

The feedback loop closes as the agent receives this reward. The RL algorithm updates the policy parameters θ to maximize the expected cumulative return. Through this iterative process, the agent learns to associate specific state-action pairs with higher energy efficiency, effectively optimizing the global metric of coverage per Joule (m^2/Joule). This allows emergent strategies, such as coasting with currents or reducing sensor polling rates in open water, to develop naturally from the reward structure.

III. Evaluation

Evaluation will be conducted in a high-fidelity ASV simulator (e.g., VRX/Gazebo or Unity) capable of modeling hydrodynamics and detailed sensor energy consumption. We define three primary test scenarios: 'Lake' (large open areas, minimal drift), 'River' (narrow channels, strong directional currents), and 'Coast' (dynamic waves, wind, dynamic obstacles). The system is trained on procedurally generated maps of each type and tested on held-out maps to ensure generalization.

We compare five experimental settings: (1) **Frontier Exploration**, a standard geometry-based baseline with sen-

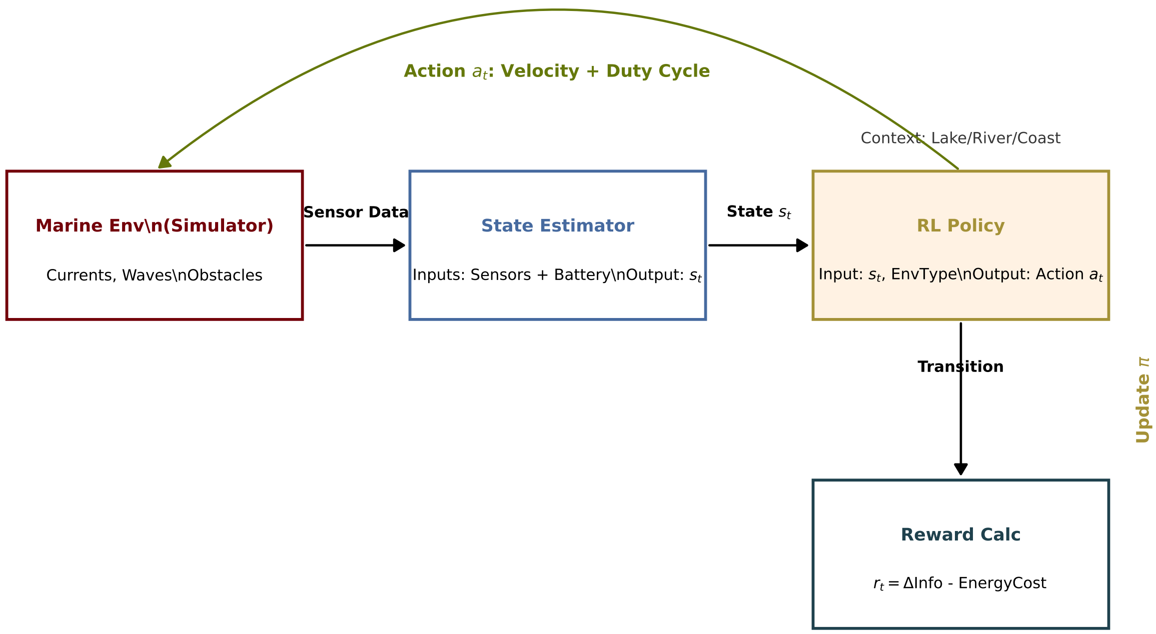

Fig. 1. Proposal 3 closed-loop method: an RL policy jointly controls vehicle motion and sensor activation, conditioned on water-body type to optimize energy efficiency.

sors always active; (2) **Rule-Based Mission Planner**, a lawnmower coverage pattern with simple battery turn-around logic; (3) **Energy-Unaware RL**, maximizing information gain without energy penalties; (4) **Energy-Aware RL (No Adapt)**, which includes energy penalties but lacks environment type input; and (5) our **Proposed Method**, which utilizes full environment adaptation and sensor duty-cycling.

The primary metrics are **Coverage Efficiency**, defined as the area mapped (m^2) per Joule of energy consumed, and **Mission Success Rate**, the percentage of missions that return to base or reach a goal before battery depletion. Secondary metrics include total information gain (entropy reduction) and safety (collision rate). We also perform stress tests, such as starting missions with critical battery levels (30%) or introducing sudden environmental shifts (e.g., transitioning from calm lake to strong river current), to evaluate robustness. Prior published results suggest that adaptive strategies significantly outperform static baselines in resource-constrained tasks (Fig. effig:published-evidence).

IV. Conclusion

This project addresses the fundamental challenge of extending the operational endurance of autonomous marine vehicles. We propose a battery-aware reinforcement learning policy that integrates sensor management with motion planning, explicitly adapting to environmental dynamics like currents and waves. By treating energy conservation as a first-class objective alongside exploration, we aim to demonstrate that intelligent, context-aware behaviors can significantly increase the effective range and utility of ASVs.

We benchmark this approach against standard frontier and rule-based methods, as well as non-adaptive RL baselines. The expected outcome is a policy that not only covers more area per Joule but also exhibits distinct, interpretable strategies for different water bodies—such as exploiting currents for transport or duty-cycling sensors in safe regions. If full RL training proves unstable, a fallback to a hierarchical approach with separate planning and power management modules will still provide valuable insights into the energy dynamics of marine autonomy.

References

[1] B. Yamauchi, “A frontier-based approach for autonomous exploration,” in *Proceedings 1997 IEEE International Symposium on Computational Intelligence in Robotics and Automation (CIRA)*. IEEE, 1997, pp. 146–151.

Comparison with Related Exploration & Energy Methods

Method	Exploration?	Energy Aware?	Env Adaptive?	Joint Control?
Frontier Exp (1997)	Yes	No	No	No
Active SLAM (2014)	Yes	Rarely	No	No
Energy-Aware Plan (201	No (Goal)	Yes	No	No
Deep RL Nav (2017)	No (Goal)	No	Implicit	No
Ours (Proposal 3)	Yes	Yes (Battery)	Yes (River/Lake)	Yes (Motion+Sensor)

Fig. 2. Related-work capability comparison and the explicit improvement axes targeted by Proposal 3.

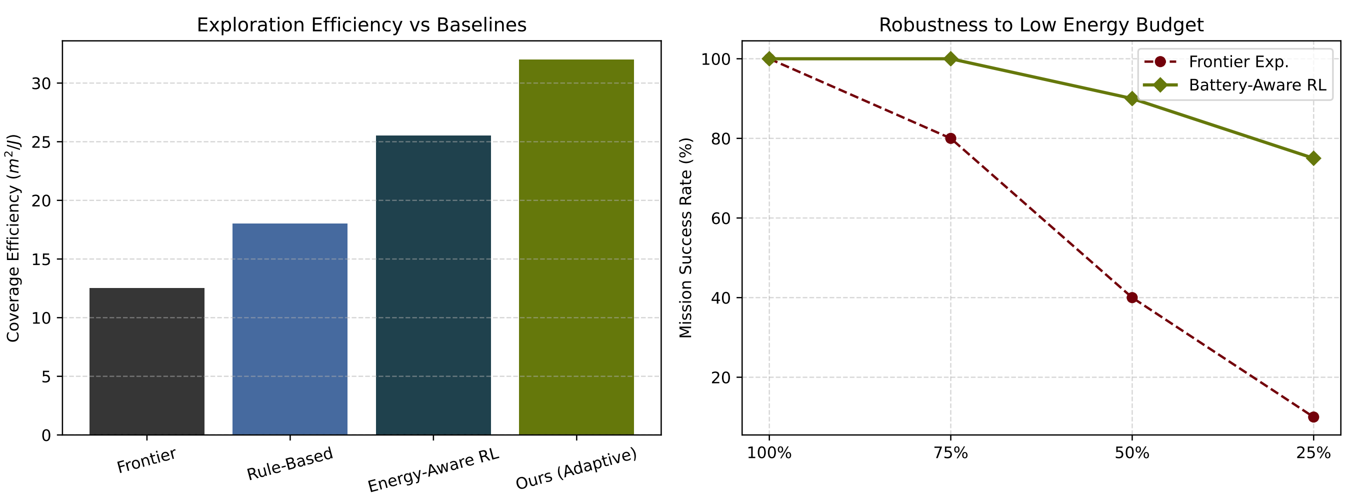

Fig. 3. Published quantitative evidence motivating Proposal 3. Left: Existing methods often separate path planning from energy management, leading to suboptimal endurance. Right: Adaptive sensing strategies significantly outperform static sensor schedules in energy-constrained environments.